

RATES OF CHANGE INTRO

06/09/25

- The following table shows the number of litres of water in my bath, W , t seconds after I opened the tap.

t	0	10	20	30	40	50	60	70	80
W	35	41	47	53	59	65	71	77	83

- What was the average rate that water flowed into the bath? Be sure to give the correct unit.
- This relationship exhibits a *constant rate*. What name do we give to such relationships? Sketch the graph (with t on the horizontal axis and W on the vertical). What is the intercept on the vertical axis? What is the gradient of the line?

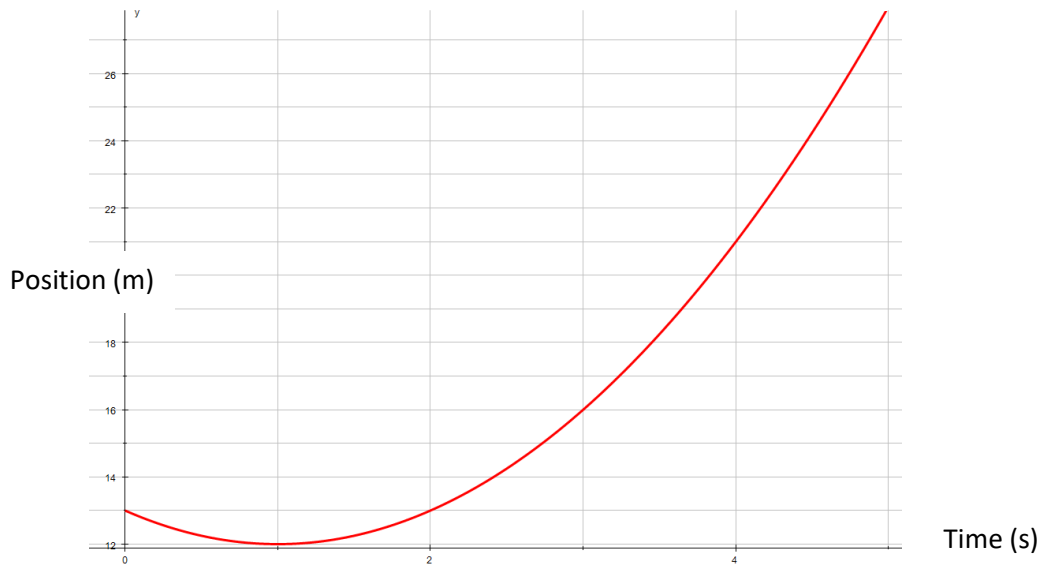
- The height of a skyscraper at various stages of its construction is shown in the table below.

days	0	30	60	90	120	150	180	210	220	225	226	228	229	230
height (m)	0	32.8	61.2	85.3	105	120	131.2	137.8	139.0	139.45	139.52	139.65	139.71	139.76

- Was the skyscraper built at a constant rate? How do you know?
- Calculate the average rate (stating the units):
 - Over the entire construction
 - Over the first 90 days
 - Over the last 80 days
 - Over the last 20 days
 - Over the last 5 days
- Find the average rate over the range of days
 - 225-230
 - 225-229
 - 225-228
 - 225-226

Hence estimate the instantaneous rate of change at day 225.

3. The following graph shows the position (in metres beyond a fixed origin) of a particle moving on along a line as a function of time.

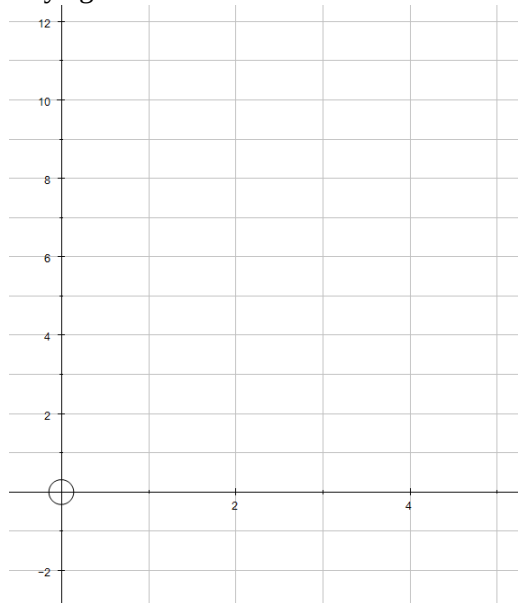


- a. With a ruler, draw tangents to the curve at 0,1,2,3,4 and 5 seconds. Measure the gradient of each tangent to estimate the *rate of change* of position at each of those times.

We give a name to the rate of change of position: *velocity*. What are the units of velocity? Complete the table below.

Time (s)	0	1	2	3	4	5
Position (m)	13					
Velocity ()		0				

- b. The rate of change of position (called velocity) could be calculated at every instant in the time interval from 0 to 5 seconds. We can therefore view the velocity itself as a function of time. Plot a graph of velocity against time on the axes below. Label the axes.



- c. The rate of change of velocity is called *acceleration*. What are the units of acceleration? Describe the acceleration in this case. Does it change over time? Is this always the case?

Rates of Change – bonus questions

1. The price of some shares was 157p at the close on 1 September and 167.5p at the close on 8 September. The share price is updated continuously by the second. Jane uses a computer to calculate the *rate of change* of the price as a function of time. She also finds the average rate over the week.

Jane says: “At some point during the week, the instantaneous rate must have been equal to the average rate.”

This sounds convincing, but Jane is wrong. Give a *counter-example* to demonstrate why.

Jane would be correct if we restrict the possible behaviours of the share price (an example of a restriction – though not the right one for this question - would be “the rate must be no higher than 10p per day”). Can you find a suitable way of restricting the behaviour of the price to make Jane correct? Can you think of a physical situation that necessarily obeys this restriction?

2. The rate of change of acceleration (or, less usefully, the rate of the rate of the rate of change of position) is called the *jerk*. What units could it be measured in?

Suppose you and your friend are passengers in two different cars. Both cars are approaching the same corner, and both drivers brake equally hard (that is, the peak acceleration during braking is the same in each car). Your driver has just passed his test and is not a smooth operator; you feel thrown around by the deceleration. The other driver is a skilled chauffeur and your friend has a smooth and comfortable ride.

Explain the differences you felt with reference to the jerk in each car. Why might it be that limiting the rate of change of acceleration gives a smoother ride? Does this restrict how hard the super-smooth chauffeur can brake/accelerate/corner?

3. Can you sketch the graph of a function for which it is *not possible* to find the rate of change at a particular moment? *Hint: what if the value of the function changes in a single instant?*

Sketch the graphs of a few functions where it is not possible to find the rate at some moment for this reason. What could they represent? What couldn't they represent?

Suppose we exclude those examples by forcing the function to be *continuous* (i.e. you can draw its graph without taking your pen off the paper). Is it still conceivable that there is some moment where it is not possible to find the rate of change? *Hint: yes.*

Sketch the graphs of a few functions where it is not possible to find the rate at some moment despite them being continuous. What features do your graphs have in common? What could they represent? What couldn't they represent?